$$x(t) = \begin{cases} 1 & |t| < T_1 \\ 0 & |t| \in (T_1, T/2) \end{cases}$$

$$a_k = \frac{\sin(k\omega_0 T_1)}{k\omega_0 T/2}$$

$$Ta_k = \frac{2\sin(\omega T_1)}{\omega} \qquad \omega = k\omega_0$$

aperiodic signal: signal with $\infty$ period

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt$$

$$\text{If: } X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$a_k = 1/T \, X(j\omega)$$

$$\Rightarrow x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T} X(jk\omega_0) e^{jk\omega_0 t}$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \omega_0 X(jk\omega_0) e^{jk\omega_0 t}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

↳ fourier transform of $x(t)$

Convergence of F.T:

→ If $x(t)$ is square integrable, then $X(j\omega)$ is finite

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$$

→ $x(t)$ has finite maxima & minimas in a finite interval

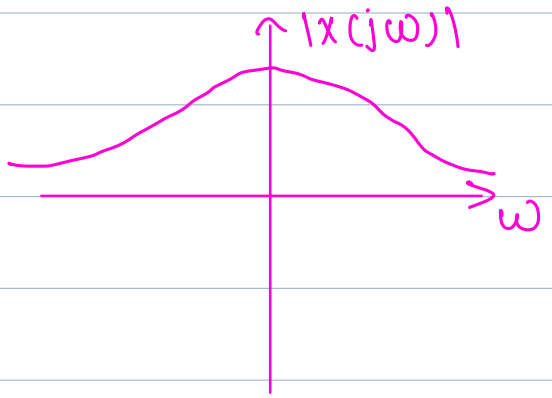
→ $x(t)$ has finite number of discontinuities within finite interval

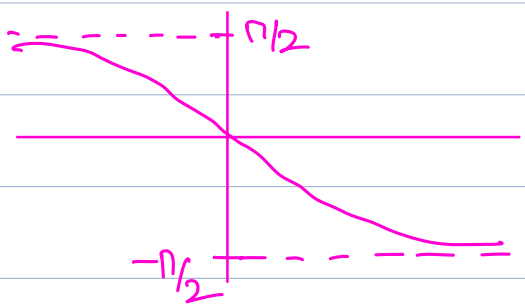
$x(t) = e^{at}u(t) \quad a>0$

$$X(j\omega) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j\omega t}dt$$

$$= \int_{0}^{\infty} e^{-(a+j\omega)t}dt = \frac{1}{a+j\omega}$$

$$|X(j\omega)| = \frac{1}{\sqrt{a^2+\omega^2}} \qquad \angle X(j\omega) = -\tan^{-1}(\omega/a)$$

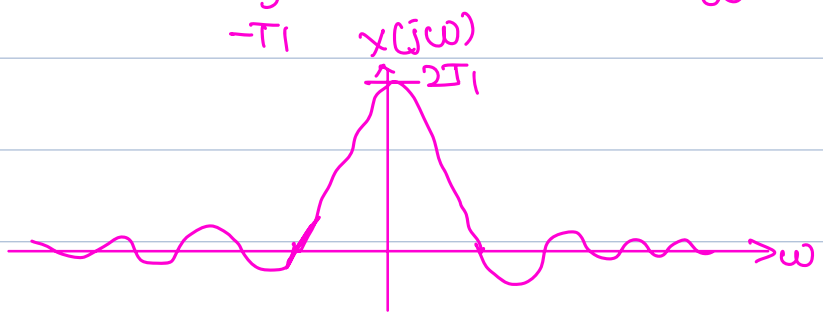



$x(t) = \delta(t) \Rightarrow X(j\omega) = 1$

$$x(t) = \begin{cases} 1 & |t| < T_1 \\ 0 & \text{else} \end{cases}$$

$$X(j\omega) = \int_{-T_1}^{T_1} e^{-j\omega t}dt = \frac{2\sin\omega T_1}{\omega}$$



$x(t) \rightarrow X(j\omega)$ = single impulse of area $2\pi$ at $\omega = \omega_0$

$$X(j\omega) = 2\pi\,\delta(\omega - \omega_0)$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega = e^{j\omega_0 t}$$

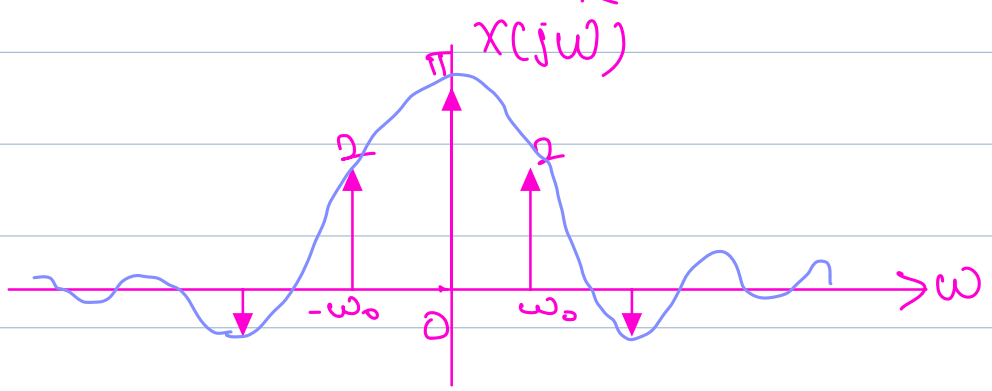
If $X(j\omega)$ is series of impulses:

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k\, \delta(\omega - k\omega_0)$$

$$\Rightarrow x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \text{F.S of } x(t)$$

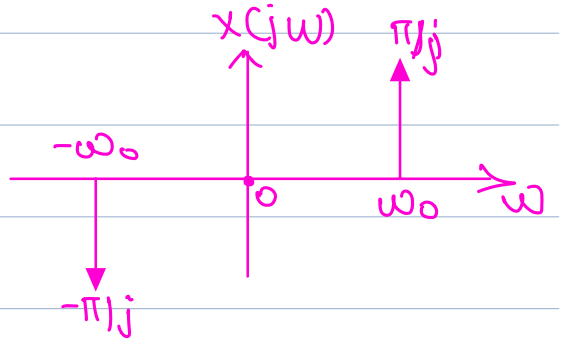
let $a_k = \dfrac{\sin(k\omega_0 T_1)}{\pi k}$

$$X(j\omega) = \sum_{k=-\infty}^{\infty} \frac{2\sin(k\omega_0 T_1)}{k}\, \delta(\omega - k\omega_0)$$



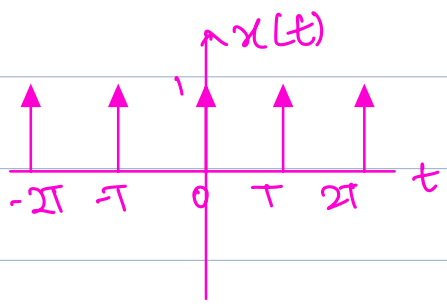
$x(t) = \sin \omega_0 t$

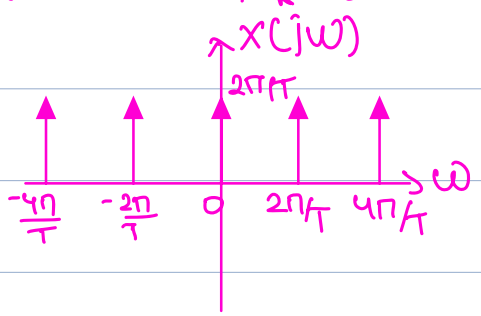
$a_1 = \frac{1}{2j} \qquad a_{-1} = -\frac{1}{2j}$




$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$

$a_k = 1/T \Rightarrow X(j\omega) = 2\pi/T \left( \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{T}) \right)$




Properties:

1) linearity: $ax(t) + by(t) \xleftrightarrow{F} aX(j\omega) + bY(j\omega)$

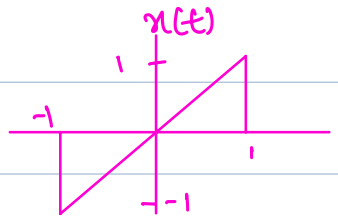
2) time shifting: $x(t - t_0) \xleftrightarrow{F} e^{-j\omega t_0} X(j\omega)$

3) conjugation: $x^*(t) \xleftrightarrow{F} X^*(-j\omega)$

$x(t)$ is real $\Rightarrow X(j\omega)$ has C.S

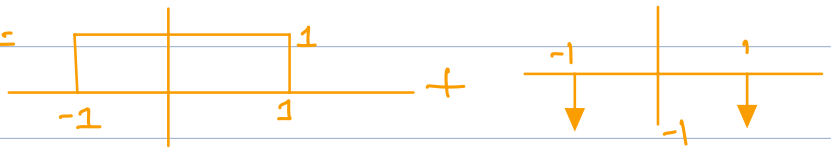
4) differentiation: $\frac{dx(t)}{dt} \xleftrightarrow{F} j\omega X(j\omega)$

5) integration: $\int_{-\infty}^{t} x(t)\,dt \xleftrightarrow{F} \frac{X(j\omega)}{j\omega} + \pi X(0)\delta(\omega)$




To calculate F.T of $x(t)$

we can calculate F.T of $g(t) = \frac{d}{dt}x(t)$

$g(t) =$ 


$$\Rightarrow G(j\omega) = \frac{2\sin\omega}{\omega} + (-e^{j\omega}) + (-e^{-j\omega})$$

$$G(0) = 0 \Rightarrow X(j\omega) = \frac{G(j\omega)}{j\omega} + \pi G(0)\delta(\omega)$$

$$= \frac{2\sin\omega}{j\omega^2} - \frac{2\cos\omega}{j\omega}$$

$X(j\omega) \rightarrow$ purely imaginary and odd

$\Rightarrow x(t) \rightarrow$ real and odd

6) $x(at) \xleftrightarrow{F} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$

7) $x(-t) \xleftrightarrow{F} X(-j\omega)$

8) $x(t)e^{j\omega_0 t} \xleftrightarrow{F} X(j(\omega-\omega_0))$

Duality: $X(j\omega)$ is F.T of $x(t)$ then

$2\pi x(-j\omega)$ if F.T of $x(jt)$

IFT of $x(j\omega)$ is $x(-t)/2\pi$

9) Convolution:

freq. resp: $H(jk\omega_0) = \int_{-\infty}^{\infty} h(t)\, e^{-jk\omega_0 t}\, dt$

$$y(t) = h(t) * x(t) \xleftrightarrow{F} Y(j\omega) = H(j\omega)\, X(j\omega)$$

10) Multiplication:

$$p(t) = x(t).y(t) \xleftrightarrow{F} p(j\omega) = \frac{1}{2\pi} X(j\omega)^{*} Y(j\omega)$$

11) Parseval's theorm:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t)\, x^{*}(t)\, dt$$

$$= \int_{-\infty}^{\infty} x(t) \left[ \frac{1}{2\pi} \int_{-\infty}^{\infty} X^{*}(j\omega)\, e^{-j\omega t}\, d\omega \right] dt$$

$$= \int_{-\infty}^{\infty} \frac{1}{2\pi} X^{*}(j\omega) \left[ \int_{-\infty}^{\infty} x(t)\, e^{j\omega t}\, dt \right] d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2\, d\omega$$

- $\lambda(\omega) = \int h(\tau) e^{-j\omega\tau} d\tau = H(j\omega)$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} \quad \text{for LTI}$$

If $x(t) = \delta(t) \Rightarrow X(j\omega) = 1$

- Invertable: non invertible if $H(j\omega)=0$ for some $\omega$
- causal: Any band-limited $H(j\omega)$ is not-causal